STA 2002, Summer 2024
Probability and Statistics II

Likelihood Ratio Test

Composite null hypothesis

- Previously, we mostly consider simple hypothesis as null hypothesis

Null Hypothesis	Alternative Hypothesis
$H_0 : \mu = \mu_0$	$H_1 : \mu \neq \mu_0$
	$H_1 : \mu > \mu_0$
	$H_1 : \mu < \mu_0$

- How to test two compositive hypothesis? For example:

$$H_0 : \mu \leq \mu_0 \qquad H_1 : \mu > \mu_0$$

Composite Hypothesis Testing

- Assume that the pdf of the random variable X is f(x;θ), where θ represents one or more unknown parameters

- We let Ω denote the total parameter space — that is, the set of all possible values of the parameter θ given by either H_0 or H_1

- Test the hypotheses

Null Hypothesis **Alternative Hypothesis**

$$H_0: \theta \in \omega, \qquad H_1: \theta \in \omega',$$

$$\Omega = \omega \cup \omega'$$

- We use the likelihood ratio as our test statistic

Reading: Chapter 8.7

Likelihood Ratio

- Assume that the pdf of the random variable X is f(x;θ), where θ represents one or more unknown parameters
- We use the likelihood ratio as our test statistic

Null Hypothesis **Alternative Hypothesis**

$$H_0: \theta \in \omega, \qquad H_1: \theta \in \omega',$$

Definition 8.7-1

The **likelihood ratio** is the quotient

$$\lambda = \frac{L(\hat{\omega})}{L(\hat{\Omega})},$$

where $L(\hat{\omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \omega$ and $L(\hat{\Omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \Omega$.

Properties of the Likelihood Ratio

Definition 8.7-1

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where $L(\hat{\omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \omega$ and $L(\hat{\Omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \Omega$.

$$H_0: \theta \in \omega, \qquad H_1: \theta \in \omega',$$

- λ is always nonnegative: $\lambda \geq 0$
- $\lambda \leq 1$ since $L(\hat{\omega}) \leq L(\hat{\Omega})$ (because $L(\hat{\omega})$ is the maximum value on a subset $\omega \subset \Omega$.)
- A small value of λ will lead to the rejection of H_0

Critical (rejection) Region

$$H_0: \theta \in \omega, \qquad H_1: \theta \in \omega',$$

- $0 \leq \lambda \leq 1$
- A small value of λ will lead to the rejection of H_0

Definition 8.7-2

To test $H_0: \theta \in \omega$ against $H_1: \theta \in \omega'$, the **critical region for the likelihood ratio test** is the set of points in the sample space for which

$$\lambda = \frac{L(\hat{\omega})}{L(\hat{\Omega})} \leq k,$$

where $0 < k < 1$ and k is selected so that the test has a desired significance level α.

Critical (rejection) Region

$$H_0: \theta \in \omega, \qquad H_1: \theta \in \omega',$$

Definition 8.7-2

To test $H_0: \theta \in \omega$ against $H_1: \theta \in \omega'$, the **critical region for the likelihood ratio test** is the set of points in the sample space for which

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where $0 < k < 1$ and k is selected so that the test has a desired significance level α.

$$-2\ln(\lambda) \sim \chi^2(\dim \Omega - \dim \omega)$$

Not required

as the sample size tends to infinity under H_0. This is true when dim Ω *and* dim ω are different, and the maximum likelihood is not taken on the boundary of the range.

Testing normal mean (variance known)

Example 8.7-1 Assume that the weight X in ounces of a "10-pound" bag of sugar is $N(\mu, 5)$. We shall test the hypothesis H_0: $\mu = 162$ against the alternative hypothesis H_1: $\mu \neq 162$. Thus, $\Omega = \{\mu : -\infty < \mu < \infty\}$ and $\omega = \{162\}$. To find the likelihood ratio, we need $L(\widehat{\omega})$ and $L(\widehat{\Omega})$. When H_0 is true, μ can take on only one value, namely, $\mu = 162$. Hence, $L(\widehat{\omega}) = L(162)$. To find $L(\widehat{\Omega})$, we must find the value of μ that maximizes $L(\mu)$. Recall that $\widehat{\mu} = \bar{x}$ is the maximum likelihood estimate of μ. Then $L(\widehat{\Omega}) = L(\bar{x})$, and the likelihood ratio $\lambda = L(\widehat{\omega})/L(\widehat{\Omega})$ is given by

$$\lambda = \frac{(10\pi)^{-n/2} \exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i - 162)^2\right]}{(10\pi)^{-n/2} \exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i - \bar{x})^2\right]}$$

$$= \frac{\exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i - \bar{x})^2 - \left(\frac{n}{10}\right)(\bar{x} - 162)^2\right]}{\exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i - \bar{x})^2\right]}$$

$$= \exp\left[-\frac{n}{10}(\bar{x} - 162)^2\right].$$

Example

A critical region for a likelihood ratio is given by $\lambda \le k$, where k is selected so that the significance level of the test is α

$$-\left(\frac{n}{10}\right)(\bar{x}-162)^2 \le \ln k,$$

$$(\bar{x}-162)^2 \ge -\left(\frac{10}{n}\right)\ln k,$$

$$\frac{|\bar{x}-162|}{\sqrt{5}/\sqrt{n}} \ge \frac{\sqrt{-(10/n)\ln k}}{\sqrt{5}/\sqrt{n}} = c.$$

$$C = \left\{\bar{x}: \frac{|\bar{x}-162|}{\sqrt{5}/\sqrt{n}} \ge z_{\alpha/2}\right\}$$

$$\lambda = \exp\left[-\frac{n}{10}(\bar{x}-162)^2\right].$$

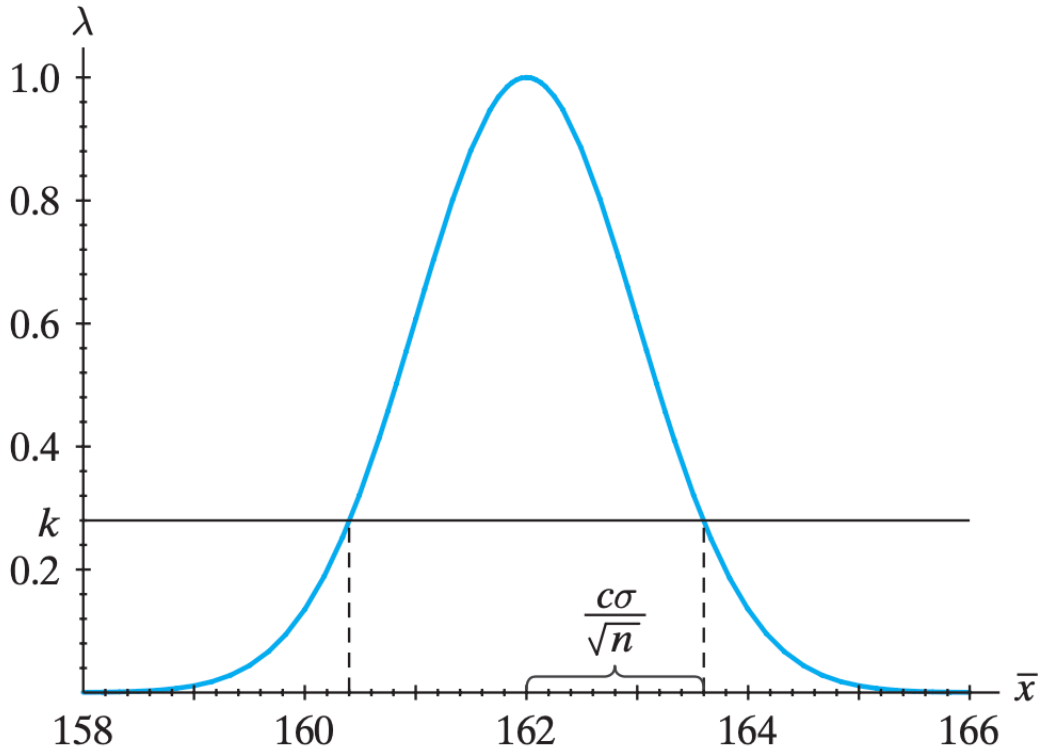


Figure 8.7-1 The likelihood ratio for testing $H_0: \mu = 162$

Testing normal mean (variance known)

- How about one-sided test? For example:

$$H_0: \mu = \mu_0 \qquad H_1: \mu > \mu_0$$

$$w = \{ \mu = \mu_0, \quad \sigma^2 = \sigma_0^2 \}$$

$$w' = \{ \mu_0 < \mu < +\infty, \quad \sigma^2 = \sigma_0^2 \}$$

$$\text{let } \Omega = w \cup w' = \{ \mu \geq \mu_0, \quad \sigma^2 = \sigma_0^2 \}$$

Testing normal mean (variance known)

$$H_0: \mu = \mu_0 \qquad H_1: \mu > \mu_0$$

under ω. $L(\mu_0) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu_0)^2}$

under Ω $L(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$

$$\ell(\mu) = \log L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Therefore, $\max_{\mu \geq \mu_0} L(\mu)$ is obtained at

$$\underset{\mu \geq \mu_0}{\arg\max}\, L(\mu) = \begin{cases} \bar{x} & \text{if } \bar{x} \geq \mu_0 \\ \mu_0 & \text{if } \bar{x} < \mu_0 \end{cases}$$

denotes the value of $\mu \geq \mu_0$ such that $L(\mu)$ is maximized.

Testing normal mean (variance known)

$$H_0: \mu = \mu_0 \qquad H_1: \mu > \mu_0$$

$$\Rightarrow \lambda = \begin{cases} \dfrac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\mu_0)^2}}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\bar{x})^2}} & \text{if } \bar{x} \geq \mu_0 \\ \dfrac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\mu_0)^2}}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\mu_0)^2}} = 1 & \text{if } \bar{x} < \mu_0 \end{cases}$$

$$= \begin{cases} e^{-\frac{n}{2\sigma_0^2}(\bar{x}-\mu_0)^2} & \text{if } \bar{x} \geq \mu_0 \\ 1 & \text{if } \bar{x} < \mu_0 \end{cases}$$

$\Rightarrow$

$\lambda \leq k \Rightarrow e^{-\frac{n}{2\sigma_0^2}(\bar{x}-\mu_0)^2} \leq k \qquad (\bar{x} \geq \mu_0)$

$\bar{x} - \mu_0 \geq \sqrt{\frac{-2\sigma_0^2 \ln k}{n}}$ (because $\bar{x} \geq \mu_0$)

$P(\lambda \leq k \mid \text{under } H_0) = \alpha$

$\Rightarrow P\left(\bar{x} - \mu_0 \geq \sqrt{\frac{-2\sigma_0^2 \ln k}{n}}\right) = \alpha.$

$\frac{\sqrt{(-2\sigma_0^2 \ln k)/n}}{\sigma_0/\sqrt{n}} = z_\alpha$

Testing normal mean (variance known)

- How about one-sided test with composite null hypothesis? For example:

$$H_0: \mu \leq \mu_0 \qquad H_1: \mu > \mu_0$$

$$\omega = \{ -\infty < \mu \leq \mu_0, \quad \sigma^2 = \sigma_0^2 \}$$

$$\omega' = \{ \mu_0 < \mu < +\infty, \quad \sigma^2 = \sigma_0^2 \}$$

$$\Omega = \omega \cup \omega' = \{ -\infty < \mu < +\infty, \quad \sigma^2 = \sigma_0^2 \}$$

Likelihood $L(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma_0^2}(x_i - \mu)^2}$

$\Rightarrow$

$$\underset{\mu \leq \mu_0}{\operatorname{argmax}} L(\mu) = \begin{cases} \bar{x} & \text{if } \bar{x} \leq \mu_0 \\ \mu_0 & \text{if } \bar{x} > \mu_0 \end{cases}$$

$$\underset{-\infty < \mu < +\infty}{\operatorname{argmax}} L(\mu) = \bar{x} \qquad \text{(the MLE)}$$

Testing normal mean (variance known)

$H_0: \mu \leq \mu_0$ $\quad$ $H_1: \mu > \mu_0$

$$\Rightarrow \lambda = \begin{cases} 1 & \text{if } \bar{x} \leq \mu_0 \\ e^{-\frac{n}{2\sigma^2}(\bar{x}-\mu_0)^2} & \text{if } \bar{x} > \mu_0 \end{cases}$$

$$\lambda \leq k \Rightarrow e^{-\frac{n}{2\sigma^2}(\bar{x}-\mu_0)^2} \leq k \qquad (\bar{x} > \mu_0)$$

$$\Rightarrow \bar{x} - \mu_0 \geq \sqrt{\frac{-2\sigma^2 \ln k}{n}} \qquad \text{since } \bar{x} > \mu_0$$

$$\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \geq \frac{\sqrt{(-2\sigma^2 \ln k)/n}}{\sigma/\sqrt{n}}$$

$$\| \\ z_\alpha$$

The same as the test for $H_0: \mu = \mu_0$ vs. $H_1: \mu > \mu_0$

Testing normal mean (variance unknown)

Suppose now that the random sample $X_1, X_2, \ldots, X_n$ arises from the normal population $N(\mu, \sigma^2)$, where both μ and σ^2 are unknown. Let us consider the likelihood ratio test of the null hypothesis H_0: $\mu = \mu_0$ against the two-sided alternative hypothesis H_1: $\mu \neq \mu_0$. For this test,

$$\omega = \{(\mu, \sigma^2): \mu = \mu_0,\ 0 < \sigma^2 < \infty\}$$

and

$$\Omega = \{(\mu, \sigma^2): -\infty < \mu < \infty,\ 0 < \sigma^2 < \infty\}.$$

Testing normal mean (variance unknown)

If $(\mu, \sigma^2) \in \Omega$, then the observed maximum likelihood estimates are $\widehat{\mu} = \bar{x}$ and $\widehat{\sigma^2} = (1/n)\sum_{i=1}^{n}(x_i - \bar{x})^2$.
Thus,

$$L(\widehat{\Omega}) = \left[\frac{1}{2\pi\left(\frac{1}{n}\right)\sum_{i=1}^{n}(x_i-\bar{x})^2}\right]^{n/2} \exp\left[-\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{\left(\frac{2}{n}\right)\sum_{i=1}^{n}(x_i-\bar{x})^2}\right]$$

$$= \left[\frac{ne^{-1}}{2\pi\sum_{i=1}^{n}(x_i-\bar{x})^2}\right]^{n/2}.$$

Similarly, if $(\mu, \sigma^2) \in \omega$, then the observed maximum likelihood estimates are $\widehat{\mu} = \mu_0$ and $\widehat{\sigma^2} = (1/n)\sum_{i=1}^{n}(x_i - \mu_0)^2$. Hence,

$$L(\widehat{\omega}) = \left[\frac{1}{2\pi\left(\frac{1}{n}\right)\sum_{i=1}^{n}(x_i-\mu_0)^2}\right]^{n/2} \exp\left[-\frac{\sum_{i=1}^{n}(x_i-\mu_0)^2}{\left(\frac{2}{n}\right)\sum_{i=1}^{n}(x_i-\mu_0)^2}\right]$$

$$= \left[\frac{ne^{-1}}{2\pi\sum_{i=1}^{n}(x_i-\mu_0)^2}\right]^{n/2}.$$

$$\Rightarrow \lambda = \frac{\left[\frac{ne^{-1}}{2\pi\sum_{i=1}^{n}(x_i-\mu_0)^2}\right]^{n/2}}{\left[\frac{ne^{-1}}{2\pi\sum_{i=1}^{n}(x_i-\bar{x})^2}\right]^{n/2}} = \left[\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\mu_0)^2}\right]^{n/2}.$$

$$= \left[\frac{1}{1+\frac{n(\bar{x}-\mu_0)^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}}\right]^{n/2}.$$

Testing normal mean (variance unknown)

$$\lambda \le k$$

$$\Rightarrow \quad \frac{1}{1+\dfrac{n(\bar{x}-\mu_0)^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}} \le k^{2/n}$$

$$\Rightarrow \quad \frac{n(\bar{x}-\mu_0)^2}{\dfrac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2} \ge (n-1)(k^{-2/n}-1).$$

$$T = \frac{\sqrt{n}(\bar{X}-\mu_0)/\sigma}{\sqrt{\sum_{i=1}^{n}(X_i-\bar{X})^2/[\sigma^2(n-1)]}} = \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sqrt{\sum_{i=1}^{n}(X_i-\bar{X})^2/(n-1)}} = \frac{\bar{X}-\mu_0}{S/\sqrt{n}} \sim t(n-1) \qquad \text{Under } H_0$$

$\Rightarrow$ That is, we reject H_0: $\mu = \mu_0$ and accept H_1: $\mu \neq \mu_0$ at the α level of significance if the observed $|T| \ge t_{\alpha/2}(n-1)$.

Testing normal variance (mean unknown)

$$\omega = \{(\mu, \sigma^2): -\infty < \mu < \infty,\ \sigma^2 = \sigma_0^2\}$$

and

$$\Omega = \{(\mu, \sigma^2): -\infty < \mu < \infty,\ 0 < \sigma^2 < \infty\}.$$

As in the test concerning the mean, we obtain

$$L(\widehat{\Omega}) = \left[\frac{ne^{-1}}{2\pi \sum_{i=1}^{n}(x_i - \bar{x})^2}\right]^{n/2}.$$

If $(\mu, \sigma^2) \in \omega$, then $\widehat{\mu} = \bar{x}$ and $\widehat{\sigma^2} = \sigma_0^2$; thus,

$$L(\widehat{\omega}) = \left(\frac{1}{2\pi\sigma_0^2}\right)^{n/2} \exp\left[-\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{2\sigma_0^2}\right].$$

$$\lambda = \left(\frac{w}{n}\right)^{n/2} \exp\left(-\frac{w}{2} + \frac{n}{2}\right) \le k,$$

$$w = \sum_{i=1}^{n}(x_i - \bar{x})^2/\sigma_0^2$$

$$\Rightarrow \quad w \le c_1 \text{ or } w \ge c_2$$